

1 One-Loop Radiative Corrections in Quantum Electrodynamics

In this chapter we shall proceed to carry out some of the classic one-loop calculations in the theory of charged leptons—massive spin $\frac{1}{2}$ particles that interact only with the electromagnetic field. There are three known species or “flavors” of leptons: the electron and muon, and the heavier, more recently discovered tauon. For definiteness we shall refer to the charged particles in our calculations here as “electrons,” though most of our calculations will apply equally to muons and tauons. After some generalities in Section 11.1, we will move on to the calculation of the vacuum polarization in Section 11.2, the anomalous magnetic moment of the electron in Section 11.3, and the electron self-energy in Section 11.4. Along the way, we will introduce a number of the mathematical techniques that prove useful in such calculations, including the use of Feynman parameters, Wick rotation, and both the dimensional regularization of ’t Hooft and Veltman and the older regularization method of Pauli and Villars. Although we shall encounter infinities, it will be seen that the final results are finite if expressed in terms of the renormalized charge and mass. In the next chapter we shall extend what we have learned here about renormalization to general theories in arbitrary orders of perturbation theory.

1.1 Counterterms

The Lagrangian density for electrons and photons is taken in the form¹

$$\mathcal{L} = -\frac{1}{4}F_B^{\mu\nu}F_{B\mu\nu} - \bar{\psi}_B[\gamma_\mu(\partial^\mu + ie_B A_B^\mu) + m_B]\psi_B, \quad (11.1.1)$$

where $F_B^{\mu\nu} \equiv \partial^\mu A_B^\nu - \partial^\nu A_B^\mu$; A_B^μ and ψ_B are the bare (i.e., unrenormalized) fields of the photon and electron, and $-e_B$ and m_B are the bare charge and mass of the electron. As described in the previous chapter, we introduce renormalized fields and charge and mass:

$$\psi \equiv Z_2^{-1/2}\psi_B, \quad (11.1.2)$$

$$A^\mu \equiv Z_3^{-1/2}A_B^\mu, \quad (11.1.3)$$

$$e \equiv Z_3^{+1/2}e_B, \quad (11.1.4)$$

$$m \equiv m_B + \delta m, \quad (11.1.5)$$

with the constants Z_2 , Z_3 , and δm adjusted so that the propagators of the renormalized fields have poles in the same position and with the same residues as the propagators of the free fields in the absence of interactions. The Lagrangian may then be written in terms of renormalized quantities, as

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_2, \quad (11.1.6)$$

where

$$\mathcal{L}_0 = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \bar{\psi}[\gamma_\mu\partial^\mu + m]\psi, \quad (11.1.7)$$

¹In this chapter we will not be making transformations between Heisenberg- and interaction-picture operators, so we shall return to a conventional notation, in which an upper case A and a lower case ψ are used to denote the photon and charged particle fields, respectively.

$$\mathcal{L}_1 = -ieA_\mu\bar{\psi}\gamma^\mu\psi, \quad (11.1.8)$$

and $\mathcal{L}_2$ is a sum of “counterterms”

$$\begin{aligned} \mathcal{L}_2 = & -\frac{1}{4}(Z_3 - 1)F^{\mu\nu}F_{\mu\nu} - (Z_2 - 1)\bar{\psi}[\gamma_\mu\partial^\mu + m]\psi \\ & + Z_2\delta m\bar{\psi}\psi - ie(Z_2 - 1)A_\mu\bar{\psi}\gamma^\mu\psi. \end{aligned} \quad (11.1.9)$$

It will turn out that all of the terms in $\mathcal{L}_2$ are of second order and higher order in e , and that these terms just suffice to cancel the ultraviolet divergences that arise from loop graphs.

1.2 Vacuum Polarization

We now begin our first calculation of a radiative correction involving loop graphs, the so-called vacuum polarization effect, consisting of the corrections to the propagator associated with an internal photon line. Vacuum polarization produces measurable shifts in the energy levels of hydrogen, and makes an important correction to the energies of muons bound in atomic orbits around heavy nuclei. Also, as we shall see in Volume II, the calculation of the vacuum polarization provides a key element in the calculation of the high energy behavior of electrodynamics and other gauge theories.

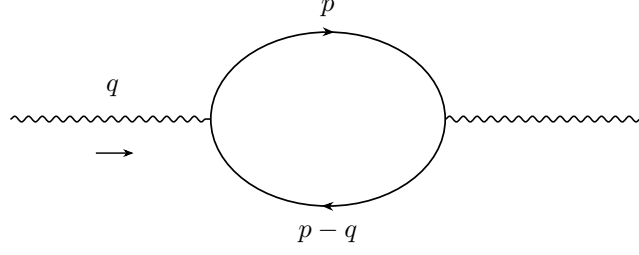


Figure 11.1. The one-loop diagram for the vacuum polarization in quantum electrodynamics. Here wavy lines represent photons; lines carrying arrows represent electrons.

As in Section 10.5, we define $i(2\pi)^4 \Pi^{*\mu\nu}(q)$ as the sum of all connected graphs with two external photon lines with polarization indices μ and ν and carrying four-momentum q into and out of the diagram, not including photon propagators for the two external lines, and with the asterisk indicating that we exclude diagrams that can be disconnected by cutting through some internal photon line. The complete photon propagator $\Delta'^{\mu\nu}(q)$ is given by Eq. (10.5.13):

$$\Delta' = \Delta [\mathbf{1} - \Pi^* \Delta]^{-1}, \quad (11.2.1)$$

where $\Delta^{\mu\nu}(q)$ is the photon propagator without radiative corrections. Our task here is to calculate the leading contributions to $\Pi^{*\mu\nu}(q)$.

In lowest order there is a one-loop contribution to Π^* , corresponding to the diagram in Figure 11.1:

$$\begin{aligned} i(2\pi)^4 \Pi_{1\text{ loop}}^{*\rho\sigma}(q) = & - \int d^4p \operatorname{Tr} \left\{ \left[\frac{-i}{(2\pi)^4} \frac{-i \not{p} + m}{p^2 + m^2 - i\epsilon} \right] [(2\pi)^4 e \gamma^\rho] \right. \\ & \times \left. \left[\frac{-i}{(2\pi)^4} \frac{-i(\not{p} - \not{q}) + m}{(p - q)^2 + m^2 - i\epsilon} \right] [(2\pi)^4 e \gamma^\sigma] \right\}. \end{aligned} \quad (11.2.2)$$

with the first minus sign on the right required by the presence of a fermion loop. More simply, this is

$$\Pi_{1\text{ loop}}^{*\rho\sigma}(q) = \frac{-ie^2}{(2\pi)^4} \int d^4p \frac{\operatorname{Tr}\{[-i \not{p} + m] \gamma^\rho [-i(\not{p} - \not{q}) + m] \gamma^\sigma\}}{(p^2 + m^2 - i\epsilon)((p - q)^2 + m^2 - i\epsilon)}. \quad (11.2.3)$$

The first step in doing this integral is to use a trick introduced by Feynman [1]. We use the elementary formula

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[(1-x)A + xB]^2} \quad (11.2.4)$$

to write the product of scalar propagators in Eq. (11.2.3) as

$$\begin{aligned}
& \frac{1}{(p^2 + m^2 - i\epsilon)((p - q)^2 + m^2 - i\epsilon)} \\
&= \int_0^1 [(p^2 + m^2 - i\epsilon)(1 - x) + ((p - q)^2 + m^2 - i\epsilon)x]^{-2} dx \\
&= \int_0^1 [p^2 + m^2 - i\epsilon - 2p \cdot qx + q^2x]^{-2} dx \\
&= \int_0^1 [(p - qx)^2 + m^2 - i\epsilon + q^2x(1 - x)]^{-2} dx.
\end{aligned}$$

(This is a special case of a class of integrals given in the Appendix to this chapter.) We can now shift the variable of integration in momentum space²

$$p \longrightarrow p + qx,$$

so that Eq. (11.2.3) becomes

$$\begin{aligned}
\Pi_{1\text{ loop}}^{*\rho\sigma}(q) &= \frac{-ie^2}{(2\pi)^4} \int_0^1 dx \int d^4p [p^2 + m^2 - i\epsilon + q^2x(1 - x)]^{-2} \\
&\quad \times \text{Tr}\{[-i(\not{p} + \not{q}x) + m]\gamma^\rho[-i(\not{p} - \not{q}(1 - x)) + m]\gamma^\sigma\}.
\end{aligned} \tag{11.2.5}$$

Using the results of the Appendix to Chapter 8, the trace here can easily be calculated as

$$\begin{aligned}
& \text{Tr}\{[-i(\not{p} + \not{q}x) + m]\gamma^\rho[-i(\not{p} - \not{q}(1 - x)) + m]\gamma^\sigma\} \\
&= 4[-(p + qx)^\rho(p - q(1 - x))^\sigma + (p + qx) \cdot (p - q(1 - x))\eta^{\rho\sigma} \\
&\quad - (p + qx)^\sigma(p - q(1 - x))^\rho + m^2\eta^{\rho\sigma}].
\end{aligned} \tag{11.2.6}$$

Our next step is called a Wick rotation [2]. As long as $-q^2 < 4m^2$, the quantity $m^2 + q^2x(1 - x)$ is positive for all x between 0 and 1, so the poles in the integrand of Eq. (11.2.5) are at $p^0 = \pm\sqrt{\mathbf{p}^2 + m^2 + q^2x(1 - x) - i\epsilon}$, i.e., just above the negative real axis and just below the positive real axis. (See Figure 11.2.) We can rotate the contour of integrations of p^0 counterclockwise without crossing either of these poles, so that instead of integrating p^0 on the real axis from $-\infty$ to $+\infty$, we integrate it on the imaginary axis from $-i\infty$ to $+i\infty$. That is, we can write $p^0 = ip^4$, with p^4 integrated over real values from $-\infty$ to $+\infty$. (If an $i\epsilon$ instead of $-i\epsilon$ had appeared in the denominator of the propagator, then we would have been setting $p^0 = -ip^4$, with p^4 again integrated over real values from $-\infty$ to $+\infty$. The effect would be a change of sign of $\Pi_{1\text{ loop}}^{*\rho\sigma}(q)$.)

²Strictly speaking, this step is only valid in convergent integrals. In principle, in order to justify the shift of variables, we should introduce some regulator scheme to make all integrals converge, such as the dimensional regularization scheme discussed below.

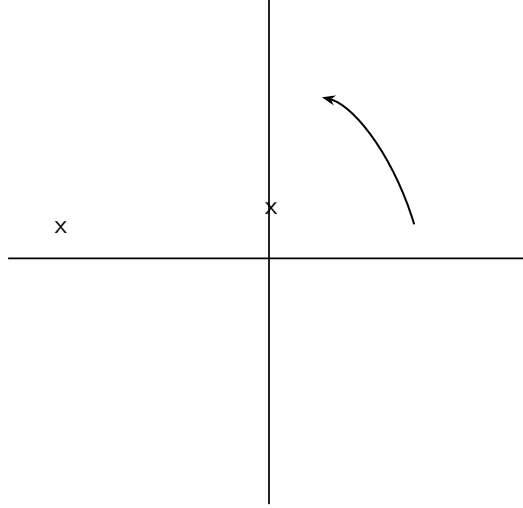


Figure 11.2. Wick rotation of the p^0 contour of integration. Small x's mark the poles in the p^0 complex plane; the arrow indicates the direction of rotation of the contour of integration, from the real to the imaginary p^0 -axes.

Equation (11.2.5) now becomes

$$\begin{aligned} \Pi_{1\text{ loop}}^{*\rho\sigma}(q) = & \frac{4e^2}{(2\pi)^4} \int_0^1 dx \int (d^4p)_E [p^2 + m^2 + q^2 x(1-x)]^{-2} \\ & \times \left[-(p+qx)^\rho (p-q(1-x))^\sigma + (p+qx) \cdot (p-q(1-x)) \eta^{\rho\sigma} \right. \\ & \left. - (p+qx)^\sigma (p-q(1-x))^\rho + m^2 \eta^{\rho\sigma} \right], \end{aligned} \quad (11.2.7)$$

where

$$(d^4p)_E = dp^1 dp^2 dp^3 dp^4$$

and all scalar products are evaluated using the Euclidean norm

$$a \cdot b = a^1 b^1 + a^2 b^2 + a^3 b^3 + a^4 b^4$$

with the understanding that $q^4 \equiv -iq^0$. Also, $\eta^{\rho\sigma}$ can be taken as either the Kronecker delta, with the indices running over 1, 2, 3, 4, or as the usual Minkowski tensor, with the indices running over 0, 1, 2, 3.

The integral (11.2.7) is badly divergent. Eventually all infinities will cancel, but to see this it is necessary at intermediate stages of the calculation to use some sort of regularization technique that makes the integrals finite. It would not do simply to cut off the integrals at some maximum momentum Λ , integrating only over p^μ with $p^2 < \Lambda^2$, because this would amount to introducing a step function $\theta(\Lambda^2 - p^2)$ into the electron propagator, and the Ward identity (10.4.25) shows that in order to maintain gauge invariance, any modification of the electron propagator must be accompanied with a modification of the electron-photon vertex. In fact, with an ordinary cutoff Λ , radiative corrections would induce a photon mass, a clear violation of the requirements of gauge invariance.

Experience has shown that the most convenient method for regulating divergent integrals without impairing gauge invariance is the dimensional regularization technique introduced by 't Hooft and Veltman [3] in 1972, based on a continuation from four to an arbitrary number d of spacetime dimensions. This amounts to carrying out angular averages in integrals like (11.2.7) by dropping all terms that are odd in p , and replacing the terms that have even numbers of p -factors with³

$$p^\mu p^\nu \longrightarrow p^2 \eta^{\mu\nu} / d, \quad (11.2.8)$$

$$p^\mu p^\nu p^\rho p^\sigma \longrightarrow (p^2)^2 [\eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}] / d(d+2). \quad (11.2.9)$$

Also, after writing the integrand in this way as a function only of p^2 , the volume element $d^4 p_E$ is to be replaced with $\Omega_d \kappa^{d-1} d\kappa$, where $\kappa \equiv \sqrt{p^2}$, and Ω_d is the area of a unit sphere in d dimensions

$$\Omega_d = 2\pi^{d/2} / \Gamma(d/2). \quad (11.2.10)$$

The integral (11.2.7) now converges for complex spacetime dimensionality d . We can continue the integral through complex d -values to $d = 4$, the infinities then reappearing as factors $(d-4)^{-1}$.

For the integral (11.2.7), dimensional regularization gives

$$\begin{aligned} \Pi_{1\text{ loop}}^{*\rho\sigma}(q) &= \frac{4e^2 \Omega_d}{(2\pi)^4} \int_0^1 dx \int_0^\infty \kappa^{d-1} d\kappa [\kappa^2 + m^2 + q^2 x(1-x)]^{-2} \\ &\quad \times \left[-\frac{2\kappa^2}{d} \eta^{\rho\sigma} + 2q^\rho q^\sigma x(1-x) + (\kappa^2 - q^2 x(1-x)) \eta^{\rho\sigma} + m^2 \eta^{\rho\sigma} \right]. \end{aligned}$$

The integrals over κ can be carried out for any complex d (or for any real d , aside from the even integers). We use the well-known formulas (given in greater generality in the Appendix to this chapter):

$$\int_0^\infty \kappa^{d-1} [\kappa^2 + v^2]^{-2} d\kappa = \frac{1}{2} (v^2)^{d/2-2} \Gamma(d/2) \Gamma(2-d/2), \quad (11.2.11)$$

$$\int_0^\infty \kappa^{d+1} [\kappa^2 + v^2]^{-2} d\kappa = \frac{1}{2} (v^2)^{d/2-1} \Gamma(1+d/2) \Gamma(1-d/2). \quad (11.2.12)$$

and find

$$\begin{aligned} \Pi_{1\text{ loop}}^{*\rho\sigma}(q) &= \frac{2e^2 \Omega_d}{(2\pi)^4} \int_0^1 dx \left[\left(1 - \frac{2}{d}\right) \eta^{\rho\sigma} (m^2 + q^2 x(1-x))^{d/2-1} \Gamma(1+d/2) \Gamma(1-d/2) \right. \\ &\quad + (2q^\rho q^\sigma x(1-x) - q^2 \eta^{\rho\sigma} x(1-x) + m^2 \eta^{\rho\sigma}) (m^2 + q^2 x(1-x))^{d/2-2} \\ &\quad \left. \times \Gamma(d/2) \Gamma(2-d/2) \right]. \end{aligned}$$

The two terms in the integrand can be combined, using

$$(1 - 2/d) \Gamma(1+d/2) \Gamma(1-d/2) = -\Gamma(d/2) \Gamma(2-d/2).$$

³These expressions may most easily be derived by noting that their form is dictated by Lorentz invariance and the symmetry among the indices μ, ν, ρ , etc., while the factors may be found by requiring that both sides give the same result when contracted with η 's.

We find

$$\begin{aligned}\Pi_{1\text{ loop}}^{*\rho\sigma}(q) &= \frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)(q^\rho q^\sigma - q^2\eta^{\rho\sigma}) \\ &\times \int_0^1 dx x(1-x)(m^2 + q^2x(1-x))^{d/2-2}.\end{aligned}\quad (11.2.13)$$

We note the very important result that this contribution to $\Pi^{*\rho\sigma}$ satisfies the relation

$$q_\rho \Pi_{1\text{ loop}}^{*\rho\sigma}(q) = 0 \quad (11.2.14)$$

that was derived in Section 10.5 on the basis of the conservation and neutrality of the electric current. It was precisely to achieve this result that we adopted the dimensional regularization scheme. The reason that dimensional regularization gives this result is that the conservation of current does not depend on the dimensionality of spacetime.

The gamma function $\Gamma(2-d/2)$ in Eq. (11.2.13) blows up for $d \rightarrow 4$. Fortunately, as we saw in Section 11.1, there is another term that must be added to $\Pi^{*\rho\sigma}(q)$, arising from the term $-\frac{1}{4}(Z_3-1)F_{\mu\nu}F^{\mu\nu}$ in the interaction Lagrangian. This term has a structure like Eq. (11.2.13)

$$\Pi_{\mathcal{L}_2}^{*\rho\sigma}(q) = -(Z_3-1)(q^2\eta^{\rho\sigma} - q^\rho q^\sigma), \quad (11.2.15)$$

so to order e^2 , the complete Π^* has the form

$$\Pi^{*\rho\sigma}(q) = (q^2\eta^{\rho\sigma} - q^\rho q^\sigma)\pi(q^2), \quad (11.2.16)$$

with

$$\pi(q^2) = -\frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)\int_0^1 dx x(1-x)(m^2 + q^2x(1-x))^{d/2-2} - (Z_3-1). \quad (11.2.17)$$

As we saw in Section 10.5, the definition of the renormalized electromagnetic field requires that $\pi(0) = 0$ (in order that the residue of the pole in the complete photon propagator at $q^2 = 0$ should be the same as for the bare propagator, aside from gauge-dependent terms). Therefore, to order e^2 ,

$$Z_3 = 1 - \frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)(m^2)^{d/2-2}\int_0^1 x(1-x)dx, \quad (11.2.18)$$

so that, to order e^2 ,

$$\begin{aligned}\pi(q^2) &= -\frac{4e^2\Omega_d}{(2\pi)^4}\Gamma(d/2)\Gamma(2-d/2)\int_0^1 dx x(1-x) \\ &\times \left[(m^2 + q^2x(1-x))^{d/2-2} - (m^2)^{d/2-2}\right].\end{aligned}\quad (11.2.19)$$

Now we can remove the regularization, allowing d to approach its physical value $d = 4$. As mentioned before, there is an infinity in the one-loop contribution, arising from the limiting behavior of the Gamma function

$$\Gamma(2-d/2) \longrightarrow \frac{1}{2-d/2} - \gamma,$$

where γ is the Euler constant, $\gamma = 0.5772157$. The infinite part of $Z_3 - 1$ is given by using $1/(2 - d/2)$ for $\Gamma(2 - d/2)$, and replacing d everywhere else by 4:

$$(Z_3 - 1)_\infty = -\frac{4e^2 \cdot 2\pi^2}{6(2\pi)^4} \frac{1}{2 - d/2} = \frac{e^2}{6\pi^2} \frac{1}{d - 4}. \quad (11.2.20)$$

We shall see in Volume II that this result may be used to derive the leading term in the renormalization group equation for the electric charge.

The poles at $d = 4$ obviously cancel in $\pi(q^2)$, because for $d = 4$ both $(m^2 + q^2x(1 - x))^{d/2-2}$ and $(m^2)^{d/2-2}$ have the same limit, unity. For the same reason, the term $-\gamma$ in $\Gamma(2 - d/2)$ cancels in the total $\pi(q^2)$, though it does make a finite contribution to $Z_3 - 1$. There are other finite contributions to $Z_3 - 1$, that arise from the product of the pole in $\Gamma(2 - d/2)$ with the linear terms in the expansion of $\Omega_d \Gamma(d/2)$ around $d = 4$, but these also cancel in the total $\pi(q^2)$. Indeed, in carrying out our dimensional regularization, we might have replaced $(2\pi)^{-4}$ with $(2\pi)^{-d}$, and the factor $\text{Tr } \mathbf{1} = 4$ might have been replaced with the dimensionality $2^{d/2}$ of gamma matrices in arbitrary even spacetime dimensionalities d , and these too would have contributed to the finite part of $Z_3 - 1$, but not of $\pi(q^2)$. Moreover, e^2 cannot be supposed to be d -independent, because as shown by inspection of Eq. (11.2.13), it has the d -dependent dimensionality $[\text{mass}]^{4-d}$. If we take $e^2 \propto \mu^{4-d}$, where μ is some quantity with the units of mass, then there are additional finite terms in $Z_3 - 1$, arising from the product of the pole in $\Gamma(2 - d/2)$ with the term $(4 - d) \ln \mu$ in the expansion of μ^{4-d} in powers of $4 - d$, but again, these cancel between $Z_3 - 1$ and the one-loop contributions to $\pi(q^2)$.

The only terms that do contribute to $\pi(q^2)$ in the limit $d \rightarrow 4$ are those arising from the product of the pole in $\Gamma(2 - d/2)$ with the linear terms in the expansion of $(m^2 + q^2x(1 - x))^{d/2-2}$ and $(m^2)^{d/2-2}$ in powers of $d - 4$:

$$(m^2 + q^2x(1 - x))^{d/2-2} - (m^2)^{d/2-2} \longrightarrow \left(\frac{d}{2} - 2\right) \ln \left(1 + \frac{q^2x(1 - x)}{m^2}\right). \quad (11.2.21)$$

This gives at last

$$\pi(q^2) = \frac{e^2}{2\pi^2} \int_0^1 x(1 - x) \ln \left(1 + \frac{q^2x(1 - x)}{m^2}\right) dx. \quad (11.2.22)$$

The physical significance of the vacuum polarization can be explored by considering its effect on the scattering of two charged particles of spin $\frac{1}{2}$. The Feynman diagrams of Figure 11.3 make contributions to the scattering S -matrix element of the form

$$\begin{aligned} S_a(1, 2 \rightarrow 1', 2') &= (2\pi)^{-12/2} \delta^4(p_{1'} + p_{2'} - p_1 - p_2) [e_1(2\pi)^4 \bar{u}_{1'} \gamma^\mu u_1] \\ &\quad \times \left[-i(2\pi)^{-4} \frac{1}{q^2} \right] [e_2(2\pi)^4 \bar{u}_{2'} \gamma_\mu u_2], \\ S_b(1, 2 \rightarrow 1', 2') &= (2\pi)^{-12/2} \delta^4(p_{1'} + p_{2'} - p_1 - p_2) [e_1(2\pi)^4 \bar{u}_{1'} \gamma^\mu u_1] \\ &\quad \times \left[-i(2\pi)^{-4} \frac{1}{q^2} \right]^2 [i(2\pi)^4 (q^2 \eta_{\mu\nu} - q_\mu q_\nu) \pi(q^2)] [e_2(2\pi)^4 \bar{u}_{2'} \gamma^\nu u_2], \end{aligned}$$

where e_1 and e_2 are the charges of the two particles being scattered; $\pi(q^2)$ is calculated using for e in Eq. (11.2.22) the magnitude of the charge of the particle circulating in the loop in Figure 11.3; and q^μ is the momentum transfer $q = p_1 - p_{1'} = p_{2'} - p_2$. Using the conservation property $q_\mu \bar{u}_{1'} \gamma^\mu u_1 = 0$ the two diagrams together yield an S -matrix element:

$$S_{a+b}(1, 2 \rightarrow 1', 2') = \frac{-ie_1 e_2}{4\pi^2 q^2} [1 + \pi(q^2)] \delta^4(p_{1'} + p_{2'} - p_1 - p_2) \times [\bar{u}_{1'} \gamma^\mu u_1] [\bar{u}_{2'} \gamma_\mu u_2]. \quad (11.2.23)$$

In the non-relativistic limit, $\bar{u}_{1'} \gamma^0 u_1 \simeq -i\delta_{\sigma_{1'}, \sigma_1}$ while $\bar{u}_{1'} \gamma^i u_1 \simeq 0$, and likewise for particle 2. Also, in this limit q^0 is negligible compared with $|\mathbf{q}|$. Equation (11.2.23) in this limit becomes

$$S_{a+b}(1, 2 \rightarrow 1', 2') = \frac{-ie_1 e_2}{4\pi^2 \mathbf{q}^2} [1 + \pi(\mathbf{q}^2)] \delta^4(p_{1'} + p_{2'} - p_1 - p_2) \delta_{\sigma_{1'}, \sigma_1} \delta_{\sigma_{2'}, \sigma_2}. \quad (11.2.24)$$

This may be compared with the S -matrix in the Born approximation due to a local spin-independent central potential $V(r)$:

$$S_{\text{Born}}(1, 2 \rightarrow 1', 2') = -2\pi i \delta(E_{1'} + E_{2'} - E_1 - E_2) T_{\text{Born}}(1, 2 \rightarrow 1', 2'). \quad (11.2.25)$$

$$T_{\text{Born}}(1, 2 \rightarrow 1', 2') = \delta_{\sigma_{1'}, \sigma_1} \delta_{\sigma_{2'}, \sigma_2} \int d^3 x_1 \int d^3 x_2 V(|\mathbf{x}_1 - \mathbf{x}_2|) \times (2\pi)^{-12/2} e^{-i\mathbf{p}_{1'} \cdot \mathbf{x}_1} e^{-i\mathbf{p}_{2'} \cdot \mathbf{x}_2} e^{i\mathbf{p}_1 \cdot \mathbf{x}_1} e^{i\mathbf{p}_2 \cdot \mathbf{x}_2}. \quad (11.2.26)$$

Setting $\mathbf{x}_1 = \mathbf{x}_2 + \mathbf{r}$, this gives

$$S_{\text{Born}} = \frac{-i}{4\pi^2} \delta^4(p_{1'} + p_{2'} - p_1 - p_2) \delta_{\sigma_{1'}, \sigma_1} \delta_{\sigma_{2'}, \sigma_2} \int d^3 r V(r) e^{-i\mathbf{q} \cdot \mathbf{r}}. \quad (11.2.27)$$

Comparing this with Eq. (11.2.23) shows that in the non-relativistic limit the diagrams of Figure 11.3 yield the same S -matrix element as a potential $V(r)$ such that

$$\int d^3 r V(r) e^{-i\mathbf{q} \cdot \mathbf{r}} = e_1 e_2 \frac{1 + \pi(\mathbf{q}^2)}{\mathbf{q}^2}$$

or, inverting the Fourier transform,

$$V(r) = \frac{e_1 e_2}{(2\pi)^3} \int d^3 q e^{i\mathbf{q} \cdot \mathbf{r}} \left[\frac{1 + \pi(\mathbf{q}^2)}{\mathbf{q}^2} \right]. \quad (11.2.28)$$

Equation (11.2.28) is to first order in the radiative correction the same potential energy that would be produced by the electrostatic interaction of two extended charge distributions $e_1 \eta(\mathbf{x})$ and $e_2 \eta(\mathbf{y})$ at a distance $\mathbf{r}$:

$$V(|\mathbf{r}|) = e_1 e_2 \int d^3 x \int d^3 y \frac{\eta(\mathbf{x}) \eta(\mathbf{y})}{4\pi |\mathbf{x} - \mathbf{y} + \mathbf{r}|}, \quad (11.2.29)$$

where

$$\eta(\mathbf{r}) = \delta^3(\mathbf{r}) + \frac{1}{2(2\pi)^3} \int d^3 q \pi(\mathbf{q}^2) e^{i\mathbf{q} \cdot \mathbf{r}}. \quad (11.2.30)$$

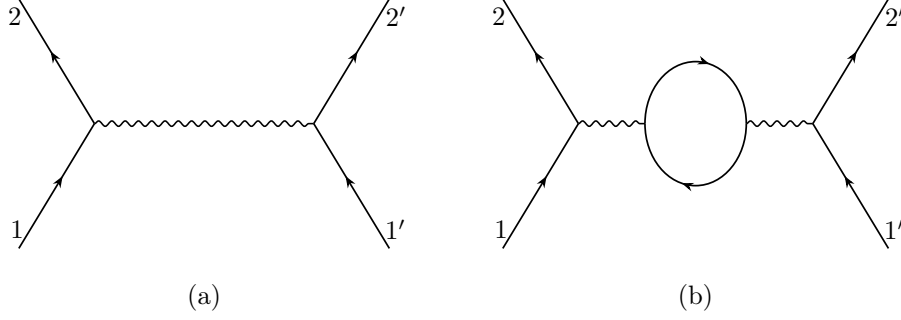


Figure 11.3. Two diagrams for the scattering of charged particles. Here lines carrying arrows are charged particles; wavy lines are photons. Diagram (b) represents the lowest-order vacuum polarization correction to the tree approximation graph (a).

Note that

$$\int d^3r \eta(\mathbf{r}) = 1 + \frac{1}{2}\pi(0) = 1, \quad (11.2.31)$$

so the total charges of particles 1 and 2, as determined from the long-range part of the Coulomb potential, are the same constants e_1 and e_2 that govern the interactions of the renormalized electromagnetic field.

For $|\mathbf{r}| \neq 0$ the integral (11.2.30) can be carried out by a straightforward contour integration:

$$\eta(\mathbf{r}) = -\frac{e^2}{8\pi^3 r^3} \int_0^1 x(1-x) dx \left[1 + \frac{mr}{\sqrt{x(1-x)}} \right] \exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right).$$

This expression is negative everywhere. However, we have seen that the integral of $\eta(\mathbf{r})$ over all $\mathbf{r}$ equals +1. Therefore, $\eta(\mathbf{r})$ must contain a term $(1+L)\delta^3(\mathbf{r})$ that is singular at $\mathbf{r} = 0$, with L chosen to satisfy Eq. (11.2.31):

$$L = \frac{e^2}{8\pi^3} \int \frac{d^3r}{r^3} \int_0^1 x(1-x) dx \left[1 + \frac{mr}{\sqrt{x(1-x)}} \right] \exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right). \quad (11.2.32)$$

The complete expression for the charge distribution function is then

$$\begin{aligned} \eta(\mathbf{r}) = (1+L)\delta^3(\mathbf{r}) - \frac{e^2}{8\pi^3 r^3} \int_0^1 x(1-x) dx \\ \times \left[1 + \frac{mr}{\sqrt{x(1-x)}} \right] \exp\left(\frac{-mr}{\sqrt{x(1-x)}}\right). \end{aligned} \quad (11.2.33)$$

The physical interpretation of this result is that a bare point charge attracts particles of charge of opposite sign out of the vacuum, repelling their antiparticles to infinity, so that the bare charge is partially shielded, yielding a renormalized charge smaller by a factor $1/(1+L)$. As a check, we may note that if we cut off the divergent integral (11.2.32) by taking the integral to extend only over $r \geq a$, we find that the part that is divergent for $a \rightarrow 0$ is

$$L_\infty = \frac{e^2}{12\pi^2} \ln a^{-1}. \quad (11.2.34)$$

Hence if we identify the momentum space cutoff Λ with a^{-1} , the divergent part of L is related to the divergent part of $Z_3 - 1$ by

$$(Z_3 - 1)_\infty = -2L_\infty, \quad (11.2.35)$$

because to order e^2 the renormalized charge (10.4.18) is given by

$$e_r = Z_3^{1/2} e_{Br} \simeq \left(1 + \frac{1}{2}(Z_3 - 1)\right) e_{Br} \simeq (1 + L)^{-1} e_{Br}. \quad (11.2.36)$$

Equation (11.2.35) is confirmed below.

Vacuum polarization has a measurable effect on muonic atomic energy levels. As we shall see in Chapter 14, the effect of Feynman graph (b) in Figure 11.3 is to shift the energy of an atomic state with wave function $\psi(\mathbf{r})$ by

$$\Delta E = \int d^3r \Delta V(\mathbf{r}) |\psi(\mathbf{r})|^2, \quad (11.2.37)$$

where $\Delta V(\mathbf{r})$ is the perturbation in the potential (11.2.28):

$$\Delta V(\mathbf{r}) = \frac{e_1 e_2}{(2\pi)^3} \int d^3q e^{i\mathbf{q}\cdot\mathbf{r}} \left[\frac{\pi(\mathbf{q}^2)}{\mathbf{q}^2} \right]. \quad (11.2.38)$$

This perturbation falls off exponentially for $r \gg m^{-1}$. On the other hand, the wave function of electrons in ordinary atoms will generally be confined within a much larger radius $a \gg m^{-1}$; for instance, for hydrogenic orbits of electrons around a nucleus of charge Ze we have $a = 137/Zm$ (where here $m = m_e$). The energy shift will then depend only on the behavior of the wave function for $r \ll a$. For orbital angular momentum ℓ , the wave function behaves like r^ℓ for $r \ll a$, so Eq. (11.2.37) gives ΔE proportional to a factor $(ma)^{-2\ell-3}$. The effect of vacuum polarization is therefore very much larger for $\ell = 0$ than for higher orbital angular momenta. For $\ell = 0$ the wave function is approximately equal to the constant $\psi(0)$ for r less than or of the order of m^{-1} , so Eq. (11.2.37) becomes

$$\Delta E = |\psi(0)|^2 \int d^3r \Delta V(\mathbf{r}). \quad (11.2.39)$$

Using Eqs. (11.2.38) and (11.2.22), the integral of the shift in the potential (for $e_1 e_2 = -Ze^2$) is

$$\begin{aligned} \int d^3r \Delta V(\mathbf{r}) &= -Ze^2 \pi'(0) \\ &= -\frac{4Z\alpha^2}{15m^2}. \end{aligned} \quad (11.2.40)$$

Also, in states of hydrogenic atoms with $\ell = 0$ and principal quantum number n the wave function at the origin is

$$\psi(0) = \frac{2}{\sqrt{4\pi}} \left(\frac{Z\alpha m}{n} \right)^{3/2}, \quad (11.2.41)$$

so the energy shift (11.2.39) is

$$\Delta E = -\frac{4Z^4\alpha^5 m}{15\pi n^3}. \quad (11.2.42)$$

For instance, in the $2s$ state of hydrogen this energy shift is -1.122×10^{-7} eV, corresponding to a frequency shift $\Delta E/2\pi\hbar$ of -37.13 MHz. This is sometimes called the Uehling effect [4]. As discussed in Chapter 1, such tiny energy shifts became measurable because in the absence of various radiative corrections the pure Dirac theory would predict exact degeneracy of the $2s$ and $2p$ states of hydrogen. As we shall see in Chapter 14, most of the $+1058$ MHz “Lamb shift” between the $2s$ and the $2p$ states comes from other radiative corrections, but the agreement between theory and experiment is good enough to verify the presence of the -37.13 MHz shift due to vacuum polarization.

Although vacuum polarization contributes only a small part of the radiative corrections in ordinary atoms, it dominates the radiative corrections in muonic atoms, in which a muon takes the place of the orbiting electron. This is because most radiative corrections give energy shifts in muonic atoms that on dimensional grounds are proportional to m_μ , while the integrated vacuum polarization energy $\int d^3r \Delta V$ due to an electron loop is still proportional to m_e^{-2} as in Eq. (11.2.40), giving an energy shift proportional to $m_\mu^3 m_e^{-2} = (210)^2 m_\mu$. However, in this case the muonic atomic radius is not much larger than the electron Compton wavelength, so the approximate result (11.2.39) only gives the order of magnitude of the energy shift due to vacuum polarization.

* * *

For the purposes of comparison with later calculations, note that if we had cut off the integral (11.2.7) at $\kappa = \Lambda$, then in place of Eq. (11.2.20) we would have encountered an integral of the form

$$(Z_3 - 1)_\infty = -\frac{e^2}{6\pi^2} \int_\mu^\Lambda \kappa^{d-5} d\kappa = \frac{e^2}{6\pi^2} \frac{\mu^{d-4} - \Lambda^{d-4}}{d-4},$$

where μ is an infrared effective cutoff of the order of the mass of the charged particle circulating in the loop of Figure 11.1. (The easiest way to find the constant factor here is to require that the limit of this expression for $d < 3$ and $\Lambda \rightarrow \infty$ matches Eq. (11.2.20).) With such an ultraviolet cutoff in place, we can pass to the limit $d \rightarrow 4$, and obtain

$$(Z_3 - 1)_\infty = -\frac{e^2}{6\pi^2} \ln(\Lambda/\mu). \quad (11.2.43)$$